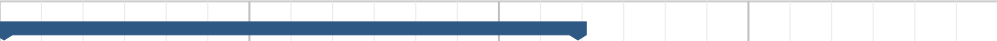


Closed-Loop Multitrophic Operational Sequencing: 120-Day Staggered Substrate Schedule

Month 1 (Days 1–30)Month 2 (Days 31–60)Month 3 (Days 61–90)Month 4 (Days 91–120)

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Primary Crop Canopy (Dwarf Wheat Chamber Batch 1)



Capillary imbibition & seed germination (Days 1–7)



Vegetative canopy development & leaf expansion (Days 8–35)



Anthesis, grain fill & transpiration recovery (Days 36–70)



Wheat Harvest & Residue Partition (Day 70)



Sintered Ceramic Substrate Regeneration (Module A)



Root severance, water purge & inspection (Day 71)



500°C pyrolytic bake-out & ECLSS gas loop recovery (Day 72)



Mineral ash flush & liquid fertilizer re-wetting (Days 73–75)



Substrate Sterile & Ready for Replanting (Day 75)



Wheat Batch 2 seeding & capillary re-start (Days 76–82)



Staggered Continuous Crop Canopy (Salad Greens)



Greens Batch 2A (Nursery to harvest, Days 10–38)



Greens Batch 2B (Nursery to harvest, Days 40–68)



Greens Batch 2C (Nursery to harvest, Days 70–98)



Saprotrophic Mycoponics (Inedible Straw Bioconversion)



Wheat straw shredding & steam pasteurization (Days 71–73)



Pleurotus ostreatus mycelial inoculation (Days 74–76)



Hyphal colonization of porous ceramic matrix (Days 77–90)



Fruiting flush induction & mushroom harvest (Days 91–98)



Microbial Bioweathering Loop (Decoupled External Reactor)



Raw regolith dust loading into isolated vessel (Days 15–17)



Acidolysis bioleaching with S. desiccabilis (Days 18–32)



Ultrafiltration & heavy metal exclusion (Days 33–35)



Purified K/Mg/P Nutrient Injection into Loop (Day 36)

